

# Gregorio Ferreira

Computational Modeling Engineer | Physics-Based FEA | Solid Mechanics | R&D Virtual Prototyping

+353-83-1373213 | gregorio.ferreira@protonmail.com | /in/gregorioferreira | Limerick City, Ireland

## Professional Summary

Computational Modeling Engineer (PhD) with nearly four years of postdoctoral and industrial R&D experience in physics-based simulation, solid mechanics, structural dynamics, and non-linear finite element analysis (FEA). Specialized in de-risking innovation and accelerating product development through virtual prototyping, automated simulation pipelines, and experimental validation. Skilled in extending commercial FEA solvers (Abaqus, Python, FORTRAN) when standard capabilities are insufficient, developing custom constitutive models, and integrating microstructural defect analysis. Proven track record of translating complex simulation results into clear, actionable design decisions for cross-functional engineering teams.

## Core Skills

**Computational FEA:** physics-based modeling; solid mechanics; structural dynamics; non-linear FEA; buckling & stability; parametric modeling

**Solver Extension:** Abaqus Implicit/Explicit; custom solver routines (UMAT/VUMAT/UEL); Python automation; MATLAB; FORTRAN; post-processing

**Material Behavior:** continuum damage mechanics; progressive failure; multiscale RVE modeling; constitutive modeling; defect-property correlation

**V&V & Testing:** model verification & validation; test-simulation correlation; full-field strain (DIC); optical microscopy; mechanical testing

**Engineering R&D:** virtual prototyping; design optimization; topology optimization; CAD/SolidWorks; DoE; R&D technical communication

**Applications:** structural components; pressure vessels; robotic manufacturing; multi-material assemblies; simulation-driven design

**Languages:** English (fluent); Portuguese (native)

## Research & Professional Experience

### Bernal Institute – University of Limerick

Sep 2022 – Present

#### Postdoctoral Researcher

Limerick, Ireland

- Develop high-predictive-capability FEA models to analyze structural stiffness, stability, and failure modes, enabling design verification and de-risking hardware investment prior to manufacturing.
- Automate simulation-driven design workflows in Python–Abaqus, linking parametric model generation, structural analysis, and automated post-processing.
- Validate numerical models against physical test data using mechanical testing, full-field strain measurement (DIC), and optical microscopy image analysis to correlate defect geometry with mechanical performance.
- Develop an open-source design-to-manufacture pipeline integrating trajectory generation, FE-based structural sizing, and automated robot code generation.
- Apply computational models to physical engineering systems (pressure vessels, tooling, and robotic cells), verifying kinematic reachability and clearance via off-line virtual cell simulation.

### HyFloatComp — Hydrogen Pressure Vessels

Apr 2025 – Present

- Simulation-led development of composite pressure vessels. Performed parametric structural analysis to optimize geometry and laminate layup before committing to physical tooling.

### VariComp — Variable-Stiffness Composite Structures

Sep 2022 – Mar 2025

- Built FE buckling and strength models for tailored structural panels and quantified defect sensitivity using microscale RVE modeling fed by optical microscopy data.

### HyCo – Hybrid Composites Technology

Jun 2021 – Aug 2022

#### R&D Mechanical Engineer

São Paulo, Brazil

- Led simulation-driven redesign of load-bearing thermoplastic and metal assemblies using 3D CAD, FEA, and topology optimization, achieving a 50% weight reduction with 15% improved structural performance.

- Carried products from initial concept to full production release through virtual prototyping, physical tooling, mechanical testing, dimensional inspection, and supplier qualification within an ISO 9001 environment.
- Translated engineering requirements into practical design specifications, authoring technical documentation and collaborating across cross-functional R&D teams.

## Research Projects — PhD Period

---

### Multiscale Modelling of Composite Material Structures

2018 – 2020

University of São Paulo

CNPq-funded · Supervisor: Prof. Volnei Tita

- Developed multiscale damage models by scripting automated FEA workflows and custom material routines. Extended standard Abaqus implicit capabilities to couple higher-order element formulations with progressive damage, validating results against tensile, bending, and impact tests.

### Damage Prediction and Monitoring in Composite Structures

2015 – 2019

University of São Paulo

CNPq-funded · Supervisor: Prof. Volnei Tita

- Applied continuum damage mechanics to predict dynamic failure and residual strength under impact. Correlated FE vibration models against measured natural frequencies and mode shapes using laser vibrometry.

## Education

---

### University of São Paulo

Oct 2014 – Nov 2019

São Paulo, Brazil

PhD in Mechanical Engineering — Aeronautical Structures

**Thesis:** Unified Finite Element Formulations for Composite Plate Structures

- Developed custom Abaqus user element (UEL) subroutines to resolve through-thickness stress profiles. Coupled UEL and UMAT subroutines within implicit FE analysis, validating model predictions against physical test campaigns.

### University of São Paulo

2012 – 2014

São Paulo, Brazil

MSc in Mechanical Engineering — Aeronautical Structures

- Implemented a progressive damage model for explicit dynamic FEA as an Abaqus VUMAT subroutine, simulating transient impact response and validating failure evolution against test data.

## Selected Publications

---

Five peer-reviewed journal papers and international conference presentations. Selected publications:

### Journal Papers

- **Ferreira, G.F.O.**, et al. "From virtual to actual assisted tape placement — application of the Frenet frame to robotic steering trajectories." *Composites Part A*, 2024. doi:10.1016/j.compositesa.2024.108369
- **Ferreira, G.F.O.**, et al. "A finite element unified formulation for composite laminates in bending considering progressive damage." *Thin-Walled Structures*, 2022. doi:10.1016/j.tws.2021.108864
- **Ferreira, G.F.O.**, et al. "Development of a finite element via Unified Formulation: implementation as a User Element subroutine to predict stress profiles in composite plates." *Thin-Walled Structures*, 2020. doi:10.1016/j.tws.2020.107107

### Selected Conference Papers

- **Ferreira, G.F.O.**, et al. "Continuous Trajectory Generation for Automated Manufacturing of Super-Ellipsoidal Composite Pressure Vessels." *ACM7*, Delft, 2026.
- **Ferreira, G.F.O.**, et al. "A Virtual Environment Based on the Frenet Frame for Manufacturing Variable Angle Tow Composite Structures with LATP." *ECCOMAS*, Lisbon, 2024.
- **Ferreira, G.F.O.**, et al. "Exploring Void Defects in LATP Manufacturing through 3D Microscale Modeling." *ICCS27*, Ravenna, 2024.

Peer reviewer for *Composite Structures* (Elsevier). Full publication list available on request.

**Work authorisation:** Irish Stamp 4 (full work rights); open to relocation to the Netherlands. **References:** on request.